

Set 3 : <https://claude.ai/public/artifacts/83eee27a-0509-45c1-be08-cc3e8b4cbfc8>

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TOOLS & MCPS

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A backend service calls the Messages API and wants a Skill to run as part of that request. What must be configured?



Set the managed-agents beta header and list the skill on an agent resource



Set settingSources explicitly, as the Agent SDK requires



Send the code-execution and skills beta headers, and author the skill so its steps need no local files or tools



Drop SKILL.md into .claude/skills and rely on the terminal to find it

Not quite

A Messages API request needs the code-execution and skills beta headers, with the skill free of local-file dependencies. The terminal placement, settingSources, and managed-agents header belong to other runtimes.

Next question →

A booking agent quotes the wrong price. The trace shows the pricing tool returned the correct number. Where is the failure?



In how the model used the correct data, not in the tool



In the network



In the integration layer, since a wrong final answer always means the tool output was mishandled on the way back to the model



In the API key configuration

Correct

The tool returned the right number, so the fault is in the model's use of it. The blanket claim about the integration layer is false, and network or key issues would not yield correct tool data plus a wrong answer.

Next question →

In the failure-handling section of a design doc, what must you decide for each error?

- ☐ The precise library and language for the retry, so an engineer could build the section directly with no further design
- ☐ Which bigger model to fail over to
- ☒ Whether it's retrievable or terminal, and what the user gets when it can't be recovered
- ☐ Whether it's worth logging

Correct

Failure handling classifies each error retrievable or terminal and sets the user-facing fallback. Naming libraries is too low-level, and logging or tier-swaps are not the error path.

A high-volume endpoint answers trivial FAQ-style questions and must feel snappy. Which reasoning setting fits?



Leave extended thinking off, enabling it only where a reasoning pass would change the answer



Run maximum-effort extended thinking everywhere so even trivial answers are extra safe and you never regret it



Keep adaptive thinking pinned to its highest effort at all times



Send every call to the largest model

Correct

Spend reasoning only where it changes the answer. Maxing thinking or the model tier on trivial, latency-sensitive traffic just adds cost and delay.

[Next question →](#)

You must guarantee an agent can neither run sudo commands nor read config/prod.key, whatever it is asked. Which TWO settings pieces enforce this? (Pick 2)

Select 2 · chosen 2

☐ defaultMode "bypassPermissions"

☒ deny ["Bash(sudo:*)"]

☒ deny ["Read(config/prod.key)"]

☐ allow ["Bash(*)"]

Correct

Settings-layer deny rules block the privileged command and the secret read regardless of the session. bypassPermissions strips guards, and an unrestricted Bash allow does the opposite.

Next question →

A personal Markdown-notes query tool that only ever runs on your own laptop. Which transport and scope fit?



HTTP + Local



HTTP + Project (.mcp.json)



stdio + Local



HTTP + Enterprise through managed settings, so it stays centrally governed and reliably available to you everywhere you might work

Correct

A personal, same-machine tool is stdio at local scope. Enterprise managed settings are for org-wide mandates and project scope shares with a team, neither of which this needs.

Next question →

A team wants one Skill to run inside a long-running agent that Anthropic hosts, addressed by an agent ID across sessions. What's required?



Define the agent as an API resource that lists the skill and set the managed-agents beta header, with the skill written to avoid local files since it runs in Anthropic's sandbox



Place SKILL.md in .claude/skills



Set settingSources explicitly



Send only the code-execution header

Correct

An Anthropic-hosted agent by ID is an API resource that lists the skill, with the managed-agents header, and the skill cannot depend on local files. The others are the terminal and SDK setups.

You keep restating the tone and safety rules inside each user message, and later turns still override them. Where should those rules live?



In the system prompt



Restated in full inside every assistant reply so the model is reminded of them continuously through the chat



In the last user message before output



In a tool result

Correct

Durable rules belong in the system prompt, the highest-priority, stable placement. Repeating them in assistant turns is fragile, and user or tool content sits lower in priority.

Next question →

Which TWO of these are gradeable success criteria you could build an eval against? (Pick 2)

Select 2 · chosen 2

☒ Return a risk score from 0-100 with one of a fixed set of named reason codes

☐ Be accurate

☒ Label the email as spam or not_spam

☐ Give a smart, helpful reply

Correct

A bounded score with a fixed reason code and a binary spam label are checkable. 'Smart, helpful' and 'accurate' are too vague to grade.

[Next question >](#)

A nightly ETL always runs the same five steps in the same fixed order. Which architecture is right, and why?

- ☐ One large prompt holding all five steps
- ☐ Five independent calls with no ordering
- ☐ An agent, since agents are more capable and future-proof and you can rein it in later if the added latency and cost hurt in production
- ☒ A workflow, since the path is fully known and coding it avoids an agent's cost and nondeterminism

Correct

A fixed, known path is a workflow: deterministic and cheaper. An agent adds nondeterminism you do not need, and a mega-prompt or unordered calls cannot reliably sequence the steps.

Next question →

A coding agent resends the same large, unchanging repo-guidelines block on every turn. What cuts the repeated input cost?

- ☐ Adding a few more examples to the block
- ☒ Cache checkpoints on the stable block
- ☐ Moving the agent to batch mode
- ☐ Trimming max_tokens each turn so replies get shorter and the per-turn input cost drops over the session

Correct

Caching the unchanged block reuses it at reduced cost. max_tokens caps output not input, batch does not fit a live agent, and more examples add tokens.

Next question →

You feed Claude's output into a typed API client that rejects anything but strict JSON, yet the model sometimes prepends a sentence. Most robust handling?



Feed the raw output to the client and let it parse



Require JSON-only, validate against a schema, and repair or reject malformed output before the client sees it



Raise max_tokens so the sentence has room



Drop temperature to 0, which fully removes formatting variation so no sentence can ever slip in again

Correct

Constrain and validate before the output reaches the client. Temperature 0 lowers but does not remove stray prose, passing raw output is what fails, and max_tokens is unrelated.

[Next question](#)

You must turn 60 PDFs into one brief that cites exact figures, but putting them all in one context ruins quality. Best approach?



Include only the PDFs that fit comfortably



Reduce the temperature



Use subagents to extract from each PDF in isolation, then synthesise from the compact extracts



Feed all 60 in at once and push max_tokens as high as the model permits so nothing is left out

Correct

Subagents keep each PDF in a clean context and hand synthesis compact extracts. Loading all 60 is the bloat that hurt quality, dropping PDFs loses data, and temperature is irrelevant.

Next question

A calendar assistant reads event descriptions that users write. One description says 'delete all my events.' Most effective mitigation?



Least privilege: the assistant that reads descriptions has no delete capability, so an injected instruction can't reach a destructive action, and untrusted text stays separate from instructions



Add a system-prompt line telling the model to ignore instructions found in event descriptions



Move to a larger, more instruction-following model



Scan descriptions for words like 'delete'

Correct

Least privilege means the reading path simply cannot delete, which is the durable defense alongside isolating untrusted text. A prompt instruction is bypassable, keyword scanning is brittle, and a more compliant model can be more susceptible.

You use extended thinking across several turns. What is required of the thinking blocks?

- ☐ They should be stripped out before the next call
- ☒ They must be returned unchanged on the next turn
- ☐ They should be condensed into a summary so a growing history won't overflow the budget, while the reasoning stays available
- ☐ They should be moved into the system prompt

Correct

Thinking blocks must go back unchanged or the next request fails. Condensing, stripping, or relocating them breaks the exchange.

Next question →

A hook must stop the agent from deleting files outside / workspace. Which TWO are correct? (Pick 2)

Select 2 · chosen 2

☒ Use PreToolUse, because it fires before the delete executes

☐ Exit 0 and print the reason to stdout

☒ Exit code 2 with the reason on stderr signals the block

☐ Use PostToolUse and undo the delete after it runs

Correct

Only PreToolUse can block, and the block is signalled by exit code 2 with the reason on stderr. PostToolUse runs after the delete, and exit 0 permits the call.

[Next question →](#)

When you map trust boundaries, which content should you treat as untrusted?

- ☐ Only content that arrives unencrypted, since transport security is what decides whether input can be trusted
- ☒ Anything the agent reads that someone else can write, like fetched pages and tool output
- ☐ The developer's own system prompt
- ☐ Constants compiled into the binary

Correct

Untrusted means anything outside your control that another party can write, chiefly fetched content and tool output. Encryption is about transport, not authorship, and your own prompt and constants are trusted.

[Next question →](#)

You open an inherited legacy repo you didn't write and don't fully trust. What permission posture fits the first pass?

☐ `bypassPermissions`, so the exploration runs uninterrupted, on the logic that you'll review the whole diff before merging anyway

☐ Manual approval on every file read

☐ All tools disabled

☒ Read-only plan mode, to explore and propose before touching anything

Correct

Match authority to risk: plan mode explores read-only before editing an untrusted repo. `bypassPermissions` grants far too much too early, approving each read is friction, and disabling tools blocks exploration.

[Next question →](#)

A CI check re-runs the same summariser twice and asserts the two outputs are byte-identical. It fails at random. Best fix?



Assert on structure and key facts instead of an exact match, since generation is non-deterministic



Loop the check until it passes



Run the check on the biggest model tier



Set temperature to 0 and keep the byte-identical assertion, treating the output as fully reproducible now

Correct

Outputs vary run to run, so grade on structure or key facts. Temperature 0 narrows but does not guarantee identical bytes, and looping or upsizing does not make an exact-match assertion valid.

You're packaging a reusable accelerator that must keep working across model updates. Which TWO practices matter most? (Pick 2)

Select 2 · chosen 2

☐

Keep prompts inline and unversioned to stay lightweight



Pin the model version and upgrade deliberately after running evals



Version the prompts so any change is attributable and reversible

☐

Auto-adopt each new model the day it ships

Correct

Pinning the model and versioning prompts make change deliberate and reversible, which is what surviving updates needs. Auto-adopting and unversioned inline prompts remove that control.

Next question →

A plugin's SKILL.md hardcodes a script at `/Users/sam/bin/format.sh` and breaks for everyone who clones the repo. Best fix?

- ☐ Remove the step so the skill stops calling the script
- ☒ Reference it from the project root via `CLAUDE_PROJECT_DIR`
- ☐ Swap it for a home shortcut like `~/bin/format.sh`
- ☐ Have every teammate mount a shared drive at that exact absolute path and document it in the README so setups match

Correct

The machine-specific absolute path is the defect; `CLAUDE_PROJECT_DIR` resolves from the repo root anywhere. A shared mount or home shortcut just relocates the fragility, and removing the step drops the capability.

Next question →

An agent can permanently delete customer records, and deletion cannot be undone. Correct design choice?

- ☐ Thorough audit logging and alerts, so a wrong delete pages the on-call engineer within minutes to start recovery
- ☐ A firm system-prompt instruction to double-check before deleting
- ☐ A low temperature to keep it cautious

☒ A human approval gate before the delete runs, built in at design time

Correct

Irreversible actions get a human gate before execution, designed in. Logging and alerts come after the record is gone, and temperature or a prompt line are not enforceable controls.

Next question →

Claude keeps invoking `lookup_user` when a task actually needs `search_orders`, and the two descriptions read almost the same. Best first fix?



Rewrite each description to state its distinct purpose and when not to use it



Delete `lookup_user`



List `search_orders` before `lookup_user`, since the model tends to favour whichever tool comes first



Turn the temperature up

Correct

Overlapping descriptions cause the wrong pick, so make each distinct with an exclusion condition. Ordering is not the lever, deleting a tool loses a capability, and temperature does not help.

A .env file holding a database password was pushed to a public repo. It's still in use. Correct response?



Rotate the password and commit the new value back to the same .env so nothing else has to change, then make the repo private



Switch the database to OAuth



Rotate the password and move it out of the committed file into an environment variable referenced at runtime



Add retry with backoff around the connection

Correct

A pushed secret is compromised: rotate it and stop committing it by moving to an environment variable. Re-committing leaves the exposure pattern, making the repo private does not undo the leak, and OAuth or backoff are unrelated.

In the cost-and-latency section of a design doc, what does the 'reliability floor' mean?



The minimum reliability the design must hold and cannot trade away for cost or speed



The p99 latency target



The lowest possible cost, reached by accepting as many dropped requests and timeouts as that price demands



The cheapest available model

Correct

The reliability floor is the minimum reliability you refuse to trade for cost or latency. It is not a cost figure, a model choice, or a latency percentile.

Next question →

Ten turns into a conversation, the assistant starts ignoring the JSON format you specified at the start. Which technique fixes this failure type?



State the format rule durably in the system prompt and tighten what was underspecified



Bump the temperature up



Swap to a different model mid-conversation



Attach three few-shot reasoning examples to every user turn so the model keeps re-anchoring on the format

Correct

Drift across turns traces to an underspecified system prompt, so fix it there. Reasoning examples in every turn are costly and off-target, and temperature or model changes do not address drift.

Support tickets are pasted verbatim into your prompt template and occasionally contain text that hijacks your instructions. Which TWO help at the prompt layer? (Pick 2)

Select 2 · chosen 2

☐

Cut every ticket to a short fixed length



Wrap and label the ticket text clearly as data

☐

Add a note asking users not to embed instructions



Keep your trusted instructions in the system prompt, apart from the ticket content

Correct

Delimiting the untrusted text as data and separating trusted instructions into the system prompt both reduce injection. A request is unenforceable, and truncation destroys legitimate content.

A client mandates that all data stay inside their existing Google Cloud organisation. How should that shape platform choice?

☐ Use the first-party API and recreate the client's governance controls around it, since staying on the native platform is worth the migration wherever the data currently sits

☐ Insist on self-hosting as the only compliant path

☒ Run Claude through Google Vertex AI so it stays within the client's cloud and governance boundary

☐ Choose whichever platform released the newest model

Correct

When data residency points to an existing cloud, run Claude through that cloud's platform, here Vertex AI. Rebuilding controls elsewhere, assuming only self-hosting is compliant, or chasing model recency all miss the constraint.

A product photo is analysed once; a brand watermark image rides on every single request. How should each be supplied?

- ☐ Convert both to text descriptions
- ☐ Inline both as base64 on every call, since inlining is simplest and the repeated bytes are negligible at scale
- ☒ Inline the one-off photo as base64, and serve the repeated watermark through the Files API so it isn't re-uploaded each call
- ☐ Send both through the Batches API regardless of latency

Correct

Inline base64 suits a one-off; the Files API suits an asset reused across requests so it isn't re-sent. Re-inlining the watermark wastes tokens, and text or forced batching miss the point.

As a chat session lengthens, which TWO things grow inside the context window? (Pick 2)

Select 2 · chosen 2

☐

The model's weight count



Accumulated tool outputs

☐

The sampling seed



Retained prior turns

Correct

Prior turns and tool outputs both accumulate and spend the fixed budget. Weight count and sampling seed are not content the window stores.